
(a) MolCoT-Paired-4K
GPT-4o-mini
Only Correct
Molecule Kept
Original CoT
Only 8% correct
Curation
Molecule Description
Curated CoT
(b) Controlled SFT
Same Training Setting
A: five base models × 10 seeds
B: six Qwen3.5 sizes × 3 seeds
0.8B
2B
4B
9B
27B
35B
(c) GRPO after SFT
Same Training Setting
Base Model
Original CoT SFT Model
Curated CoT SFT Model
GRPO
reward
(d) Evaluation
Molecule Correctness:
EM, RDK-FTS, Validity
CoT Correctness